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Clinical Characteristics and Natural History of Branch Retinal Artery Ischemia

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Objective: To describe the clinical features and natural history of branch retinal artery ischemia (BRAI)—an uncommon condition where chronic, partial branch retinal artery obstruction (also known as partial BRAO) by a non-occlusive arterial plaque causes sectoral retinal ischemia and hemorrhages.

Design: A retrospective, multicenter cohort study.

Subjects: Thirteen patients diagnosed with BRAI between January 1, 2013, and December 31, 2023.

Methods: We identified patients with BRAI based on the presence of (1) a retinal arterial plaque, (2) sectoral round hemorrhages in the distribution of the affected artery, and (3) delayed arterial flow on fluorescein angiography (FA). We reviewed longitudinal medical records, color fundus images, FA, OCT, and OCT angiography images where available. We also systematically reviewed images of patients diagnosed with Hollenhorst plaque, BRAO or central retinal artery occlusion, or ocular ischemic syndrome to estimate BRAI prevalence.

Main Outcome Measures: Clinical features, visual acuity (VA), and central subfield thickness.

Results: We identified 13 patients with BRAI with a median follow-up of 1.7 years (range 0.1–6.3 years). The mean age was 74.2 ± 9.4 years, with 61.5% men. Most patients had a history of hypertension (76.9%), hyperlipidemia (84.6%), carotid artery disease (69.2%), or type 2 diabetes mellitus (69.2%). Most patients (53.9%) maintained good VA (20/25 or better) over the years, and none developed macular edema or neovascularization during follow-up. The retinal hemorrhages fluctuated in severity over time but did not correlate with VA or central subfield thickness.

Conclusions: Branch retinal artery ischemia is defined by a triad of features: retinal arterial plaque, sectoral retinal hemorrhages, and delayed arterial flow. Most patients with BRAI have systemic vascular risk factors but maintain good vision without complications over years.

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Retinal vascular events present with a myriad of clinical manifestations depending on the location, acuity, and severity of the event. Sudden, total obstruction of retinal arteries causes acute retinal ischemia and leads to branch or central retinal artery occlusion (BRAO or CRAO). These retinal artery occlusions may cause sudden vision loss accompanied by whitened, opacified retinal edema, "box-carring" of blood vessels, and attenuation of the retinal arteries on fundus examination, with progressive retinal thinning over time. Central retinal artery occlusion is considered to be a stroke equivalent. Although practice patterns vary by geography, thrombolysis may be effective in visual recovery if administered within the stroke window.[1] Risk factors for CRAO and BRAO include factors that contribute to atherosclerosis and elevated intraocular pressure, such as arterial hypertension, ischemic cardiac disease, peripheral artery disease, history of stroke, renal disease, cataract, and glaucoma.[2]

Chronic hypoperfusion of the retina may also occur as a result of carotid artery stenosis, leading to ocular ischemic syndrome (OIS). Ocular ischemic syndrome is characterized by retinal arterial narrowing and mid-peripheral round retinal hemorrhages on fundus examination and delayed arterial filling on fluorescein angiography (FA). The degree of vision loss is variable and typically gradual. Chronic ischemia due to OIS can also result in complications such as retinal neovascularization and neovascular glaucoma, which may result in permanent vision loss or even loss of the eye.[3–5]

Unlike BRAO or OIS, branch retinal artery ischemia (BRAI) was first described by Yiu and Mukkamala in 2018 as a chronic, sectoral hypoperfusion state caused by a partially occlusive Hollenhorst plaque.[6] This uncommon clinical presentation is characterized by the presence of a retinal arterial plaque accompanied by sectoral round retinal hemorrhages and delayed arterial filling in the vascular distribution distal to the plaque. Unlike OIS, the retinal hemorrhages in BRAI are only present in a sectoral distribution along the affected retinal artery. The hemorrhages are small and round, unlike the flame-shaped hemorrhages that more often occur in retinal vein occlusions (RVOs). In contrast to retinal artery occlusions, there

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is no retinal whitening or progressive thinning. Similar phenomena such as "partial" or "incomplete" BRAO have been reported, which may lead to paracentral acute middle maculopathy.[7] Here, we performed a cohort study of patients diagnosed with BRAI to better characterize its natural history and conducted a systematic image review of patients diagnosed with related retinal vascular conditions to estimate the prevalence of this clinical entity.

Methods

Patient Selection

We conducted a retrospective, multicenter cohort study of patients diagnosed with BRAI contributed by retinal specialists at the University of California, Davis (Davis, CA), Mayo Clinic Health System (Rochester, MN), Advanced Retina (Milwaukee, WI), and Massachusetts Eye & Ear Infirmary (Boston, MA). The requirement for informed consent was waived because of the retrospective nature of the study. Patients were confirmed to have BRAI if they met the following criteria of (1) presence of a retinal arterial plaque, (2) sectoral, round retinal hemorrhages in the distribution of the affected artery, and (3) delayed arterial filling on FA.[6] We distinguished Hollenhorst plaques by their refractile, yellow, crystal-shaped appearance, which differs from the dull, white or gray, and nonrefractile appearance of platelet-fibrin or calcific emboli. We also distinguished round retinal hemorrhages from the flame-shaped hemorrhages more often seen in RVOs and accompanied by cotton-wool spots.

We additionally performed a systematic review of clinical records and ophthalmic images from all patients diagnosed with BRAO (*International Classification of Diseases* [ICD]9 code 362.32 or ICD10 codes H34.239, H34.233, H34.232, H34.231, and H34.239), OIS (ICD10 code H35.82), or Hollenhorst plaque (ICD9 code 362.33 or ICD10 codes H34.219, H34.213, H34.212, and H34.211) at the University of California, Davis and the Mayo Clinic who had fundus images available for review. Additional patients identified from this systematic review who met the same criteria were also included in the cohort analysis. All included cases were reviewed by $\geq$2 ophthalmology board-certified retina specialists (R.I., R.V., K.M., and G.Y.). This retrospective study was approved by the Institutional Review Board (Protocol # 992141) at the University of California, Davis. The study was conducted in accordance with ethical principles outlined in the Declaration of Helsinki.

Clinical Chart and Image Review

For patients diagnosed or identified with BRAI, we reviewed medical records to collect indication for appointment, age at diagnosis, sex, tobacco use, and medical history or comorbidities, including carotid artery disease, embolism, stroke, hypercoagulability state or thrombosis, hypertension, hyperlipidemia, cardiac arrhythmias, myocardial infarction, diabetes mellitus, anemia, cancer, renal disease, and cardiac surgery. Medical history was documented by medical professionals and retrieved using electronic health records. We also reviewed ocular characteristics, including eye laterality, other ocular conditions, lens status, best-recorded visual acuity (VA) at the time of diagnosis and at yearly intervals, and any vision loss events, defined as a decrease in the VA from baseline by $\geq$2 lines at any visit.

We reviewed retinal images, including ultra-widefield or color fundus photo images, FA, OCT, and OCT angiography when available. Fundus autofluorescence images were occasionally used when the shape or pattern of retinal hemorrhages could be better discerned on this modality.

Statistics

The mean age, VA, and central subfield thickness were reported as mean $\pm$ standard deviation, whereas gender, lens status, and medical or ocular comorbidities were presented as proportions. Change in VA and central subfield thickness over time was assessed by linear regression analysis with time as an independent variable.

Results

Clinical Characteristics of BRAI

Of the 13 patients who met the definition of BRAI, the mean age was 74.2 $\pm$ 9.4 years, and 61.5% were males. Most cases of BRAI were diagnosed incidentally on follow-up for comorbid ocular conditions (61.5%) or as incidental findings on routine eye examinations (38.5%). Most of the patients had a history of hypertension (76.9%), hyperlipidemia (84.6%), carotid artery disease (69.2%), and type 2 diabetes mellitus (69.2%) at the time of diagnosis. Among the patients with carotid artery disease and ultrasound imaging, a majority had bilateral disease (77.8%) or a diseased carotid ipsilateral to the BRAI eye (22.2%; Table S2, available at www.ophthalmologyscience.org). Patient 8 did not have ultrasound imaging available for review but had documentation of a prior left carotid endarterectomy. None of the patients had a prior history of amaurosis fugax and reported such symptoms during follow-up. Some had a history of prior stroke (30.8%) or were current or former tobacco users (46.2%). A proportion of patients had a history of mild nonproliferative diabetic retinopathy without macular edema (38.5%), presence of epiretinal membrane (30.8%), or glaucoma or ocular hypertension (23.1%). Most patients (84.6%) were asymptomatic at presentation (Table 1). Mean baseline VA was logarithm of the minimum angle of resolution (logMAR) 0.15 $\pm$ 0.21 (Snellen equivalent 20/29), with more than half of patients (53.9%) having Snellen VA of 20/25 or better at presentation (Table 1). Notable exceptions include patient 9, who had a presenting VA of 20/50 that was likely caused by punctate epithelial erosions and dry eye syndrome with a baseline VA of 20/40 before the appearance of a Hollenhorst plaque, and patient 12, who had a presenting VA of 20/80 that was attributed to long-standing macular edema of unclear etiology that resolved before presentation (Table 1). Most patients did not undergo visual field testing, and only patients 2 and 5 underwent automated Humphrey visual field testing for glaucoma monitoring, which were both reliable with nonspecific defects correlating to the distribution of the BRAI.

Consistent with the selection criteria, eyes with BRAI show sectoral round hemorrhages in the arterial distribution distal to a Hollenhorst plaque (Figs 1A, E). Most of the retinal arterial plaques were located in the superotemporal quadrant (38.5%) and the inferotemporal quadrant (38.5%), followed by the inferonasal quadrant (15.3%), and only 1 plaque was located in the superonasal quadrant (7.7%). The plaques were located mostly in first-order (61.5%), followed by second-order (30.8%) and third-order (7.7%) arterioles. Fluorescein angiography images

Table 1. Clinical and Ocular Characteristics of Patients with BRAI

Patient Number	Age (Yrs)	Sex	Past Medical History	Past Ocular History	Indication for Examination	Eye Laterality	Hollenhorst Plaque Location	Imaging Modality	Symptoms	VA at Baseline Visit	VA at Final Visit	Follow-Up Duration (Yrs)
1	80	Male	CAD, CVA, CABG, CA, DM2, HL, HTN	Mild NPDR	NPDR follow-up	Right	Inferotemporal	OCT, CFP, FA, FAF, OCTA	Absent	20/20	20/20	6
2	75	Female	CAD, embolism, CVA, CA, anemia, DM2, HL, HTN	Mild NPDR, glaucoma suspect	Glaucoma follow-up	Left	Inferonasal	OCT, CFP, FA, FAF, OCTA	Absent	20/30	20/40	6
3	78	Male	CAD, MI, arrhythmias, DM2, HL, HTN	Mild NPDR	Postcataract follow-up	Right	Inferotemporal	OCT, CFP, FA	Absent	20/15	20/20	3
4	86	Male	CAD, MI, arrhythmias, HL, HTN	Anterior uveitis, CME, glaucoma suspect	Anterior uveitis follow-up	Right	Superotemporal	OCT, CFP, FA	Absent	20/40	20/40	6
5	78	Male	CAD, CABG, CVA, CA, arrhythmias, DM2, HL, HTN	Mild NPDR, glaucoma, ERM	ERM follow-up	Left	Superotemporal Inferotemporal Inferonasal	OCT, CFP, FA	Present	20/40	20/50	6
6	71	Female	CAD, DM2	Mild NPDR, vitreomacular traction	NPDR follow-up	Left	Superotemporal	OCT, CFP, FA	Absent	20/40	20/50	4
7	68	Male	CAD, MI, CA, anemia, arrhythmias, CKD, HL, HTN	None	Incidental findings	Left	Superonasal	OCT, CFP, FA, FAF	Absent	20/20	20/25	1
8	57	Male	CAD, DM2, HL, HTN	None	Floaters	Right	Inferotemporal	OCT, CFP, FA	Absent	20/20	20/20	2
9	86	Female	CAD, CVA, arrhythmias, anemia, thrombosis, CA, CKD, DM2, HL, HTN	None	Annual diabetic eye examination	Right	Superotemporal	OCT, CFP, FAF	Absent	20/50	20/40	1
10	56	Male	DM2	None	Incidental findings	Right	Superotemporal	OCT, CFP, FA	Absent	20/20	20/20	<1
11	69	Female	HL	ERM	Incidental findings	Right	Inferonasal	OCT, UWF-FA, UWFI	Present	20/20	20/20	<1
12	85	Female	CAD, DM2, HL, HTN	ERM	Incidental findings	Right	Inferotemporal	OCT, UWFI	Absent	20/80	20/100	<1
13	76	Male	HL, HTN	None	Incidental findings	Left	Superotemporal	OCT, UWF-FA, UWFI	Absent	20/20	20/20	<1

BRAI = branch retinal artery ischemia; CA = cancer; CABG = coronary artery bypass grafting; CAD = carotid artery disease; CFP = color fundus photo; CKD = chronic kidney disease; CME = cystoid macular edema; CVA = cerebrovascular accident; DM2 = diabetes mellitus type II; ERM = epiretinal membrane; FA = fluorescein angiography; FAF = fundus autofluorescence; HL = hyperlipidemia; HTN = hypertension; MI = myocardial infarction; NPDR = nonproliferative diabetic retinopathy; OCTA = OCT angiography; UWF-FA = ultra-widefield fluorescein angiography; UWFI = ultra-widefield imaging; VA = visual acuity.

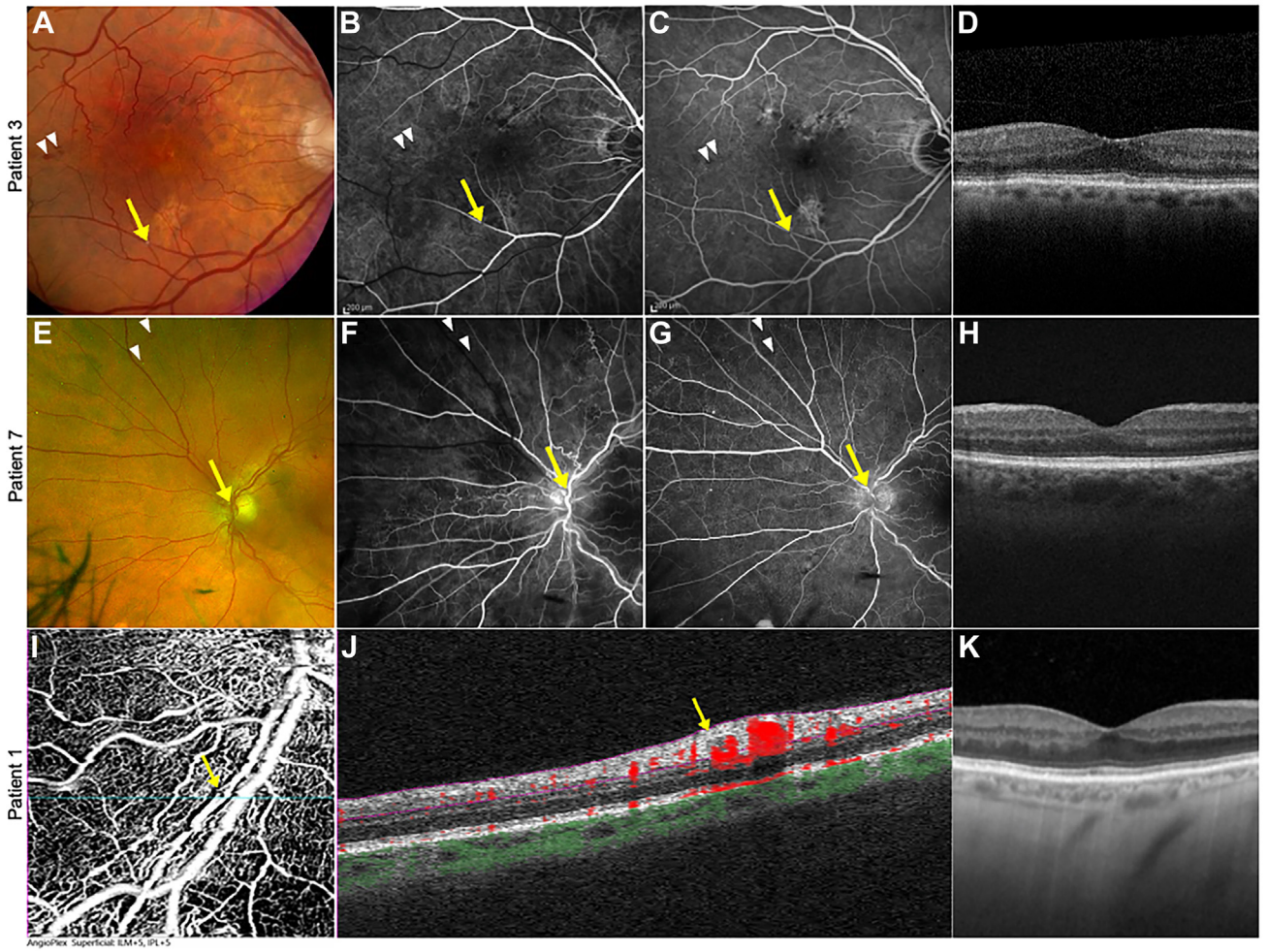


Figure 1. Clinical and imaging features of BRAI. Ultra-widefield fundus images showing a Hollenhorst plaque (arrow) and round retinal hemorrhages (arrowheads) in patient 3 **(A)** and patient 7 **(E)**. Early- and late-phase fluorescein angiogram images showing partial arterial obstruction by the plaque and delayed arterial filling in patient 3 **(B)** and patient 7 **(F)**. Macular OCT images showing absent retinal thinning or edema in patient 3 **(D)**, patient 7 **(H),** and patient 1 **(K)**. OCT angiography **(I)** and € structural OCT images **(J)** in patient 1 showing obstructed flow around the Hollenhorst plaque on OCT angiography, with no retinal thinning seen. BRAI = branch retinal artery ischemia.

show delayed filling distal to the plaque in early arterial or arteriovenous phases (Figs 1B, F), but complete filling at late phases (Figs 1C, G). Macular OCT showed no evidence of macular edema, retinal thinning, or paracentral acute middle maculopathy, likely because all of the cases occurred outside the macular region (Figs 1D, H and K). In the few cases where OCT angiography or OCT images were captured from the retinal region where the obstruction or hemorrhages occurred, flow signal could be detected around but not within the Hollenhorst plaque, with no clear retinal thinning observed distal to the lesion (Figs 1I–J).

Natural History of BRAI

Over a median follow-up duration of 1.7 years (range: 0.1–6.3 years), eyes with BRAI maintained stable vision without treatment, with a mean VA of logMAR 0.19 ± 0.15 (Snellen equivalent 20/31) at year 1, logMAR 0.2 ± 0.21 (Snellen equivalent 20/32) at year 2, and logMAR 0.27 ± 0.22 (Snellen equivalent 20/37) at year 3 (Fig 2A). Eight of the 9 patients who had ≥1 years of follow-up maintained stable VA during the follow-up period (Fig 2A). Only 1 patient developed vision loss at year 3, from VA 20/40 to 20/80, presumably due to retinal atrophy from previously treated macular edema that was not attributed to the BRAI by the retinal provider. Central subfield thickness as measured on OCT remained stable for most patients, except for patient 5, who had an epiretinal membrane without macular edema at presentation and later underwent pars plana vitrectomy with membrane peeling at year 3 (Fig 2B).

Longitudinal review of fundus images showed that the retinal hemorrhages fluctuate in severity over years, occasionally increasing or decreasing in number and size (Fig 3). However, all of the retinal arterial plaques observed in our study remained in their location at the time of diagnosis and did not move further downstream nor resolve during

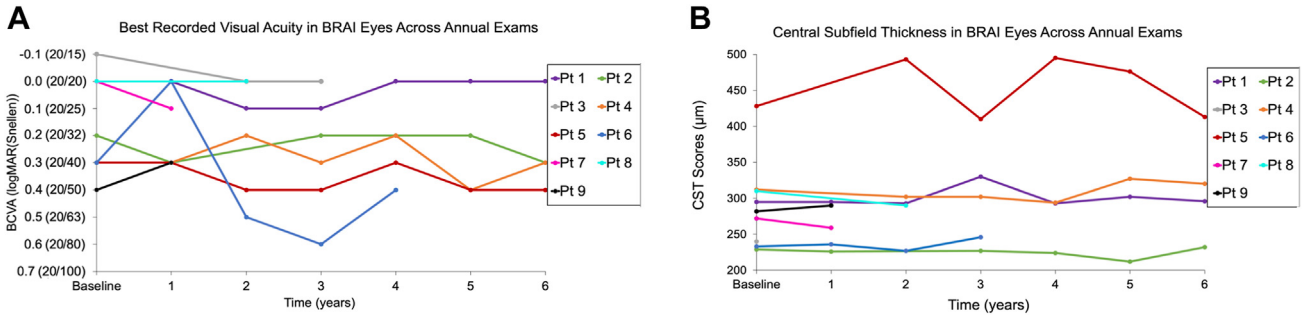


Figure 2. Natural history of BRAI. Line plots showing **(A)** best-recorded VA change (logMAR [Snellen]) and **(B)** CST measurements at yearly visits of patients with BRAI with ≥1 years of follow-up. BCVA = best-corrected visual acuity; BRAI = branch retinal artery ischemia; CST = central subfield thickness; logMAR = logarithm of the minimum angle of resolution; VA = visual acuity.

follow-up. Due to inconsistent availability of general health records, it was difficult to determine if these fluctuations may have been associated with blood pressure control or cardiovascular status. Patients who developed decreased vision during the follow-up period did not show any apparent association with the hemorrhage severity. None of the patients developed other retinal occlusive events such as BRAO, branch RVO, CRAO, or ischemic stroke during the follow-up period.

Estimated Prevalence of BRAI

To estimate the prevalence of BRAI, we systematically reviewed retinal images from 124 patients diagnosed with BRAO, OIS, or Hollenhorst plaque who had images available for evaluation (Table S1, available at www.ophthalmologyscience.org). We identified 7 cases of BRAI from 92 patients diagnosed with BRAO, no cases of BRAI among patients with OIS, and 1 BRAI case from 11 patients with Hollenhorst plaques.

Discussion

Whereas retinal artery occlusions such as BRAOs and CRAOs are typically caused by complete occlusion of the arterial lumen by a plaque or embolus, resulting in acute ischemia and hypoxic retinal injury, partial or incomplete obstruction of retinal vessels can lead to chronic ischemia with features of OIS but in a sectoral pattern in the distribution of the obstructed arteriole. We described this uncommon clinical entity as a BRAI, which is characterized by a clinical triad of a retinal arterial plaque, sectoral retinal hemorrhages, and delayed arterial filling in the distribution of the affected retinal artery or arteriole. In this study, we report the natural history of a small cohort of BRAI patients from 4 clinical practices from geographically distinct regions with follow-up of up to 6 years. We found that BRAI typically presents asymptomatically with good VA and, despite the waxing and waning nature of the retinal hemorrhages, often remains stable for years without progression or vision loss. Many patients with BRAI had vascular comorbidities such as hypertension, hyperlipidemia, diabetes, prior carotid artery disease, and prior stroke, similar to risk factors for other ischemic retinal arterial diseases such as OIS, CRAO, and BRAO.[2]

The excellent VA at presentation and asymptomatic nature of the condition are expected, as most of the retinal pathology in BRAI occurs outside the macular region and does not appear to be associated with neovascularization or macular edema. These characteristics are similar to those of BRAO, which also presents typically with preserved central acuity and rarely leads to neovascularization or macular edema, although roughly 50% of patients with BRAO develop permanent visual field deficits.[8,9] Our BRAI cases are distinguished from BRAO by the presence of retinal hemorrhages and absence of retinal whitening or thinning, although we cannot exclude the possibility that BRAI may be an impending stage before BRAO or a reperfused stage after BRAO in some cases. Nevertheless, the distinct clinical characteristics and outcomes of the patients in our study suggest that BRAI should be carefully distinguished as a separate condition from BRAO.

Unlike BRAO, 16% of patients with OIS develop ocular neovascularization, with 64% developing neovascularization of the iris and 71% developing neovascularization of the angle within a mean time to diagnosis of neovascularization of 82 days (range 22−268 days).[10] Ocular neovascularization may lead to permanent vision loss and requires treatment with intravitreal anti-VEGF injections, pan-retinal photocoagulation, or other surgical interventions.[5] No patients with BRAI developed ocular neovascularization during our study period.

Most retinal arterial plaques in BRAI cases occurred in first-order arteries, which is expected as smaller arterioles are more likely to become fully occluded, although the factors that may predispose an embolus to become partially obstructed in BRAI or fully occlusive to cause BRAO are not clear. No patients with BRAI went on to develop additional vascular occlusive events such as BRAO, branch RVO, CRAO, or ischemic stroke within the cumulative 34.7 person-years of follow-up in our study. In contrast, the 3-year stroke incidence is 36% after CRAO and 30% after BRAO.[11] Nonetheless, the prevalence of pre-existing vascular and atherosclerotic disease in patients with BRAI was high. Therefore, an isolated finding of BRAI features in a patient without known vascular risk factors should prompt

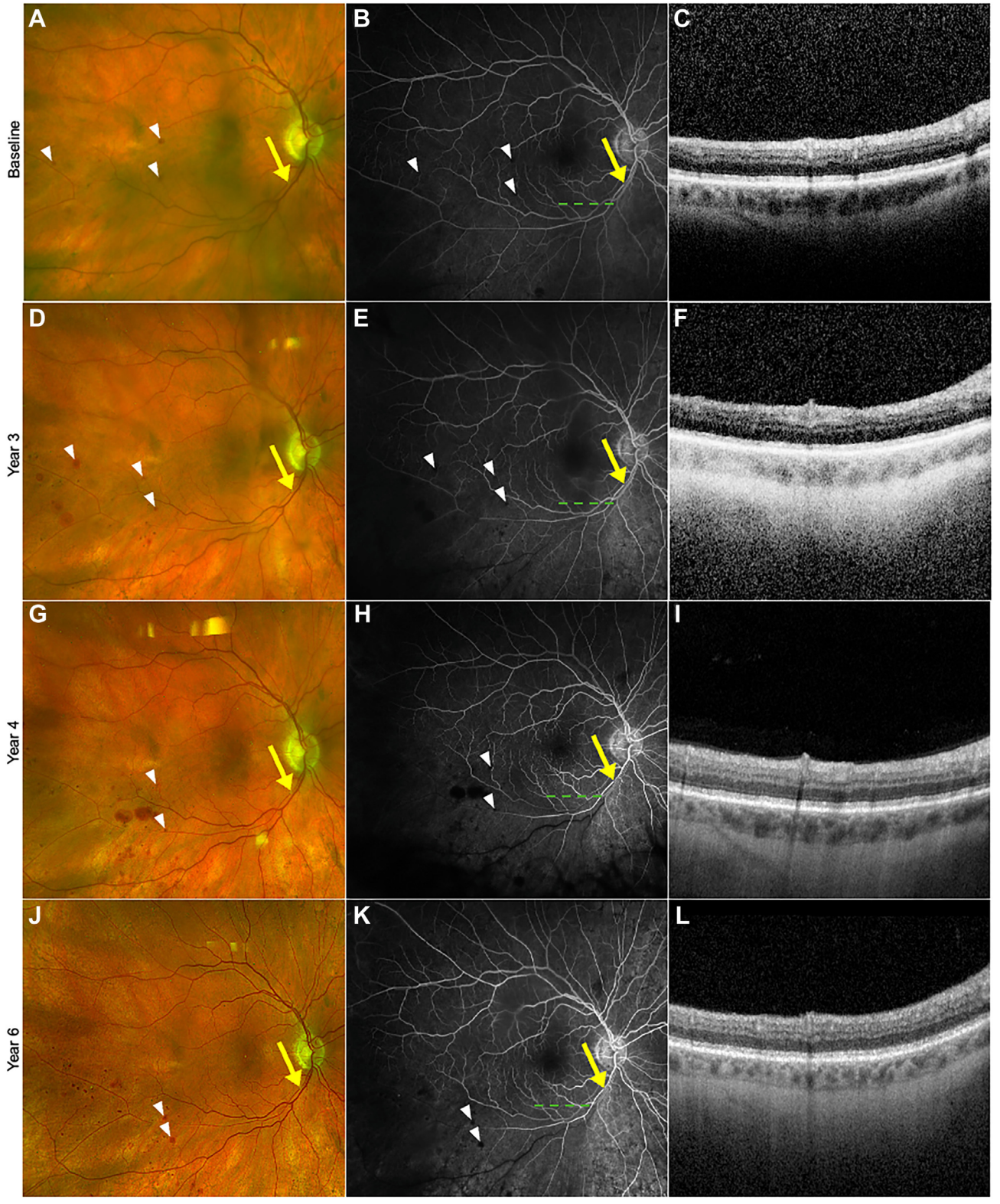


Figure 3. Longitudinal variations in BRAI clinical appearance. Ultra-widefield fundus images **(A, D, G,** and **J)**, fluorescein angiograms **(B, E, H,** and **K),** and OCT images **(C, F, I,** and **L)** of patient 1 over 6 years duration, showing persistent delay in arterial flow but fluctuations in retinal hemorrhage appearance without impacting visual acuity. Hollenhorst plaques are depicted by a yellow arrow, hemorrhages by arrowheads, and corresponding locations of OCT images by a green dashed line. BRAI = branch retinal artery ischemia.

referral for evaluation of vascular, cardiac, and renal diseases.

We hypothesize that a similar type of chronic hypoperfusion and ischemia occurs due to central retinal artery stenosis. Some patients with OIS features have no identifiable carotid or ophthalmic artery stenosis, which may be caused by a partial occlusion of the central retinal artery leading to chronic ischemia and the OIS phenotype on fundus examination.[12,13] This, along with BRAI and OIS, may represent a spectrum of retinal vascular disease due to partial obstruction and chronic hypoperfusion of the carotid, ophthalmic, central, and branch retinal arteries.

The limitations of this study include its retrospective nature, small cohort size, and lack of long-term longitudinal follow-up with serial imaging and examinations for all patients. Greater awareness is necessary to identify more patients to more fully characterize the natural history and comorbidities associated with BRAI. Based on our careful review of images from patients previously diagnosed with BRAO, OIS, or Hollenhorst plaques, we estimated that the prevalence of BRAI is between 7.6% and 9.1% in patients with these conditions, although the actual prevalence may be higher as not all patients had all available imaging modalities available for review. Also, because all patients are from US regions, its generalizability to other populations outside the United States is unknown.

In conclusion, BRAI is a chronic sectoral retinal arterial hypoperfusion state that is comprised of a triad of clinical findings: (1) retinal arterial plaque, (2) sectoral retinal hemorrhages, and (3) delayed arterial filling on FA. Most patients with BRAI are asymptomatic and maintain good VA over years without developing neovascularization, macular edema, or additional vaso-occlusive events such as ischemic stroke, although a majority of patients with BRAI have multiple pre-existing vascular risk factors. The presence of sectoral hemorrhages on fundus examination should prompt close evaluation for a retinal arterial plaque and consideration of BRAI as a diagnosis.

Footnotes and Disclosures

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G.Y.: Consultant − 4DMT, AbbVie, Adverum, Alcon, Bausch & Lomb, Boehringer Ingelheim, Clearside, Coave, Genentech, Janssen, jCyte, Mobius, Novartis, Ocuphire, Opthea, Ray Therapeutics, RegenXBio, Samsung Bioepis, Stealth Therapeutics, Supra Medical, Surrozen, West Pharmaceuticals; Honoraria − Bausch & Lomb; Travel expenses − Adverum; Other financial interest − Bausch & Lomb (food and beverage).

HUMAN SUBJECTS: Human subjects were included in this study. The requirement for informed consent was waived because of the retrospective nature of the study. This retrospective study was approved by the Institutional Review Board (Protocol # 992141) at the University of California, Davis. All research adhered to the tenets of the Declaration of Helsinki.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Connor, D'Souza, Iezzi, Yiu

Data collection: Connor, D'Souza, Iezzi, Vogel, Al-Moujahed, Miller, Moussa, Yiu

Analysis and interpretation: Connor, D'Souza, Iezzi, Yiu

Obtained funding: Yiu

Overall responsibility: Connor, D'Souza, Iezzi, Vogel, Yiu

Abbreviations and Acronyms:
BRAI = branch retinal artery ischemia; **BRAO** = branch retinal artery obstruction; **CRAO** = central retinal artery occlusion; **FA** = fluorescein angiography; **ICD** = *International Classification of Diseases*; **logMAR** = logarithm of the minimum angle of resolution; **OIS** = ocular ischemic syndrome; **RVO** = retinal vein occlusion; **VA** = visual acuity.

Keywords:
Branch retinal artery ischemia, Hollenhorst plaque, Ocular ischemia, Retinal artery occlusion, Retinal hemorrhage.

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